

TRUST CHAIN · OPERATOR-SIGNED, NOTHING LISTED ON THE SERVER

- O Operator**: the single trust root
signs every account claim
one signing key, held apart
- A Account claim**: one per isolation boundary
a signed token, not config
minted by the control plane
- U User credential**: one holder, one credential
named for the process
a browser gets a token

operator-signed claims define each account

WHERE THE CLAIMS LIVE

- R Account resolver**: signed tokens on disk
no account list in
the server config file
- C Claim push**: over the system API
add or revoke live
no server restart

served when a client connects

ACCOUNT PLANE · ONE ACCOUNT PER ISOLATION BOUNDARY

- S System account**: server and claim administration
one holder: the accounts service
carries no business traffic
- P Platform account**: shared control-plane services
reference data, registry, traces
one account every tenant shares
- T Tenant account**: the hard isolation boundary
own streams, key-value, files
cannot see another tenant
- N Tenant account, next**: added while the platform runs
claim minted, pushed, then used
no restart, no config edit

every subject is published inside one account

SUBJECT FAMILIES · THE VOCABULARY THE GRANTS ARE WRITTEN IN

- E evt.**: durable domain facts
evt.{context}.{service}.
stays inside its account
- R rpc.**: service to service
rpc.{context}.{svc}.{op}.v1
crosses only by a grant
- A api.**: browser to service
api.{context}.{svc}.{op}.v1
never given to a backend
- N notify.**: change notification
notify.{context}.{what}
replaces server-sent events
- O obs.**: debug side channel
obs.trace. and obs.pubsub.
never on a business path

only the subjects a grant names may cross

CROSS-ACCOUNT CONTRACT · THE ONLY PATH BETWEEN TWO ACCOUNTS

- D Reference data, down**: the platform account exports, each tenant imports
the tenant calls a plain local subject; the remap is hidden
one shared corpus, not a copy for every tenant
- U Telemetry and stream views, up**: each tenant exports, the platform account imports
the server writes the tenant name into the subject
read-only views; a client cannot forge the name
- X Everything else is refused**: no grant means no delivery, and no clue to guess from
create, update, purge and delete are never exported
no tenant can name another tenant's subject

! **Isolation is enforced by the server, not by application code**
signed claims · one account per boundary · explicit export and import · subject-safe identifiers only
a subject token never carries the tenant or the region: the account carries the tenant, the deployment carries the region

Read it downward. The operator issues identity, the resolver holds it, the account plane spends it, the subject families name what is spoken, and the contract row is the only opening in the wall. Everything drawn here runs today in the Dictionary POC with four accounts and live tenant creation. **Not shown:** individual subjects, permission claim text and stream settings — those are L4.

WORDS ON THIS SHEET	
ATS the messaging system (a product name)	creds a credentials file: claim plus key
WT JSON Web Token, a signed claim	grant an export matched by an import
PI application programming interface	subject the address a message is sent to
PC remote procedure call	remap the server rewriting a subject
OC proof of concept (the Dictionary POC)	{context} company or business unit token
claim a signed statement of who you are	{tenant} the account name the server inserts

EDGE · LEAF NODES AT A SITE (PROPOSED, NOT BUILT TODAY)

- D Depot leaf node**: a small server at the site connects out to its region proposed · not built today
- F Fleet gateway leaf**: telematics and device feeds tidied at the site, then sent proposed · not built today
- C Site credential**: the site holds its own claim its subjects are site-limited proposed · not built today
- W Why a leaf, not a client**: the site works when the link drops it buffers, then replays one wide-area link per site

one leaf connection out to the cell

INSIDE ONE REGIONAL CELL · THE SAME SHAPE REPEATS IN EVERY REGION

- N Three NATS servers**: one cluster, links fully meshed clients 4222, health 8222 three is a minimum, not a size
- J JetStream store**: files on disk, one domain per cell a domain is a cell, not a tenant one copy in lab, three proposed
- B Web socket entry**: browsers, through the web server a short-lived token, no file proven in the Dictionary POC
- M Monitoring endpoint**: server, link and stream health read by the operator console not a public endpoint

published messages are kept as durable state

DURABLE STATE, HELD PER ACCOUNT

- S Shipping log**: the aggregate log evt.*.shipping kept so it can replay
- R Refdata feed**: changes, aged out evt.*.refdata platform account
- O Observability**: traces and messages one hour, capped dropped under load
- K Key-value**: derived, rebuildable ships, container, meta a bucket is a stream
- F Object store**: evidence documents organizations-docs big files stay out

core subjects cross an allowed gateway

named streams copied by mirror or source

BETWEEN CELLS · EVERY TILE HERE IS PROPOSED AND NOT BUILT TODAY

- G Gateway**: joins two clusters, without stretching one carries only the core subjects a list allows proposed · not built today
- M Mirror or source**: copies one named stream into another cell there is no cross-region copy by default proposed · not built today
- T One operator, one resolver**: every cell trusts the same signing authority an account keeps one identity in every cell proposed · not built today

FAILURE BEHAVIOUR · WHAT HAPPENS WHEN A LINK BREAKS

- 1 Client reconnect**: a service never gives up retries are set to unlimited the default stops after sixty
- 2 Gateway partition**: each cell serves its own clients core messages do not queue proposed · not yet tested
- 3 Leaf disconnect**: the site buffers, then replays order is kept per subject proposed · not yet tested
- 4 Consumer restart**: it resumes at its last mark the kept log supplies the gap proven in the Dictionary POC

WORDS ON THIS SHEET	
NATS the messaging system (a product name)	domain a JetStream name marking one cell
JetStream the part of NATS that stores messages	stream a kept, replayable log of messages
HA high availability, keeping running through a fault	consumer a named reading position on a stream
cell one region's complete runtime	mirror / source a copy of one stream into another
cluster servers linked as one, in one place	ack a reader confirming it handled a message
gateway a link joining two clusters	
leaf node a small server at a site, linked to a cell	

- **Purple boundary** – the edge, away from a data centre
- **Grey boundary** – behaviour, not a component
- **Amber dashed line** – a message kept as durable state

-  **Cyan boundary** — the runtime inside one region
-  **Solid tile** — built and running in the lab today
-  **Cyan dotted line** — a copy of a stream, not the original

-  **Green boundary** — durable state that survives a restart
-  **Dashed tile** — proposed for V3, not built today
-  **Arrow head** — the direction of the relationship

-  **Amber boundary** — links between two regions
-  **Blue line** — a live connection between servers
- Line label** — names the relationship and its transport